MULTICOMMODITY FLOW, THE RELAXATION TIME AND THE DIAMETER

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ABSTRACT. For a finite reversible Markov chain on n states let $\inf_{\mathbf{f}} \psi(\mathbf{f})$ be the least congestion of a family of unit flows from each state to the stationary distribution, in the normalisation of Aldous and Fill; we call it the flow parameter. Sinclair showed that it is at most a constant times the relaxation time τ_2 times $\log n$. Open Problem 4.49(ii) of Aldous and Fill's *Reversible Markov Chains and Random Walks on Graphs*, which they attribute to Sinclair, asks whether $\log n$ can be replaced by the diameter Δ of the transition graph. For general chains the answer is negative for a simple reason: adding transitions of small probability between all pairs of states of the random walk on an expander makes $\Delta = 1$, does not increase τ_2, and keeps the flow parameter of order $\log n$. We show that the answer is negative also for simple random walks on graphs, for every constant and already on graphs of diameter two. For the simple random walk on the cone over a d-regular Ramanujan graph on n vertices the diameter is 2, the relaxation time is at most $(d+1)/(d+1-2\sqrt{d-1})$, and the flow parameter tends to $2(d+1)$ as $n \to \infty$; so the ratio of the flow parameter to $\tau_2\Delta$ is at least $d+1-2\sqrt{d-1}$ in the limit and is unbounded over d. The length-weighted form of the flow parameter fails as well, on these graphs and already on bounded-degree expanders. On the cones over the incidence graphs of projective planes of order s the flow parameter has a closed form, and the ratio increases with s to $5/2$ and exceeds 2 exactly when $s \geq 38$. For simple random walks a diameter bound does hold with a constant of order $\log D$, D the maximum degree, and the cones show that this order cannot be lowered below $\log D / \log\log D$.

1. INTRODUCTION

Let P be an irreducible reversible transition matrix on a finite state space V of size n with stationary distribution π, and let $1 = \lambda_1 > \lambda_2 \geq \cdots \geq \lambda_n \geq -1$ be its eigenvalues. The relaxation time is $\tau_2 = (1-\lambda_2)^{-1}$. Following [1, Chapter 3], a flow is an antisymmetric function $f_{ij} = -f_{ji}$ on ordered pairs, supported on the pairs with $p_{ij} > 0$; its net outflow at i is $f_{(i)} = \sum_j f_{ij}$. A unit flow from a to π is a flow with $f_{(i)} = 1_{(i=a)} - \pi_i$ for every i [1, (3.98)]. For a family $\mathbf{f} = (\mathbf{f}^{(a)})_{a \in V}$ of unit flows from each a to π, Lemma 4.21 of [1] considers the quantity

$$\psi(\mathbf{f}) = \max_{(i,j)} \sum_{a \in V} \pi_a \frac{|f_{ij}^{(a)}|}{\pi_i p_{ij}}, \tag{1}$$

the maximum over the edges (i,j) of the transition graph, and proves that the flow induced by the chain itself satisfies $\psi \leq \tau_1^{(4)} \leq \Delta\, \psi$, where Δ is the diameter of the transition graph and $\tau_1^{(4)}$ is one of the mixing-time parameters of Chapter 4. We call $\inf_{\mathbf{f}} \psi(\mathbf{f})$ the flow parameter.

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Aldous and Fill do not name ψ, and in their Section 4.5 the name flow parameter is used for

$$\tau_c = \sup_A \frac{\pi(A)\pi(A^c)}{Q(A,A^c)}, \qquad Q(A,A^c) = \sum_{i\in A,\, j\notin A} \pi_i p_{ij},$$

the supremum over nonempty proper subsets A of V; we call τ_c the cut parameter. Sinclair [15] uses the corresponding multicommodity-flow quantity, the resistance ρ of the chain, and $\rho \le \inf_{\mathbf{f}} \psi(\mathbf{f}) \le 2\rho$, as one sees by decomposing flows into paths. In the notes to Section 4.3.5, Aldous and Fill write that Sinclair [15] deduced from the multicommodity flow theorem of Leighton and Rao [9] that

$$\inf_{\mathbf{f}} \psi(\mathbf{f}) \le K\, \tau_2 \log n \tag{2}$$

for a universal constant K, and that Sinclair posed the following.

> Open Problem 4.49 of [1]. (i) Is there a simple proof of (2) in general? (ii) Does (2) hold with the diameter Δ in place of $\log n$?

For general chains the answer to (ii) is negative for a simple reason (Remark 2.2): adding transitions of small probability between all pairs of states of the random walk on an expander makes the diameter 1, does not increase the relaxation time, and keeps the flow parameter of order $\log n$. A simple random walk has no transitions of small probability to add, and for simple random walks the answer is also negative, in a strong form: no constant works in the diameter form $\inf_{\mathbf{f}} \psi(\mathbf{f}) \le K\tau_2\Delta$ of (2), already on graphs of diameter two.

Theorem 1.1. *For every $K < \infty$ there is a finite connected simple graph G of diameter 2 such that the simple random walk on G satisfies*

$$\inf_{\mathbf{f}} \psi(\mathbf{f}) > K\, \tau_2\, \Delta.$$

Consequently $\sup_G \inf_{\mathbf{f}} \psi(\mathbf{f})/(\tau_2\Delta) = \infty$, the supremum over simple random walks on finite connected graphs, so the diameter form of (2) fails for every universal constant even within this class.

The graphs are cones: a hub joined to every vertex of a d-regular expander. On such a cone the hub keeps the diameter at 2 and the expansion of the base keeps the relaxation time bounded, but every base vertex has a single edge to the hub and d edges into a base whose typical distances are large, so almost all of the flow has to pass through the spokes, and the congestion on a spoke is close to $2(d+1)$. The statement with the parameters made explicit is the following; $\mu_2(H)$ denotes the second largest adjacency eigenvalue of H.

Theorem 1.2. *Let $d \ge 3$ and let H be a connected d-regular graph on $n \ge d+2$ vertices. For the simple random walk on the cone $G = K_1 \vee H$,*

$$\Delta(G) = 2, \qquad \tau_2(G) = \frac{d+1}{d+1-\mu_2(H)},$$

and, with $B_r = d\big((d-1)^{r-1}-1\big)/(d-2)$, for every integer $r \ge 1$

$$2(d+1)\,\frac{r}{r+d}\Big(1-\frac{1+B_r}{n}\Big) \;\le\; \inf_{\mathbf{f}} \psi(\mathbf{f}) \;\le\; 2(d+1)\Big(1-\frac{d+1}{n(d+2)}\Big). \tag{3}$$

In particular $\inf_{\mathbf{f}} \psi(\mathbf{f}) \to 2(d+1)$ along any sequence of connected d-regular graphs H with $n \to \infty$, and if $\mu_2(H) \le \beta$ along the sequence for some $\beta < d+1$ then

$$\liminf_{n\to\infty} \frac{\inf_{\mathbf{f}} \psi(\mathbf{f})}{\tau_2\, \Delta} \;\ge\; d+1-\beta.$$

Ramanujan graphs [10, 12, 11] have $\mu_2 \le 2\sqrt{d-1}$, and by the Alon–Boppana bound [13] no d-regular family does better, so within cones over regular graphs the limiting ratio is at most $d+1-2\sqrt{d-1}$ and Ramanujan bases attain it. This is 1.17 for $d=3$, 2 for $d=5$, 5 for $d=10$ and 10 for $d=17$; it tends to infinity with d, which is Theorem 1.1. Random regular graphs give the same conclusion through Friedman's theorem [6, 2].

For one family the flow parameter can be computed exactly. On the cone over the incidence graph of a projective plane of order s an explicit family of flows attains $\inf_{\mathbf{f}} \psi(\mathbf{f})$: it sends part of the traffic between the most distant pairs of base vertices through the hub and spreads the rest evenly over their shortest paths in the base (Proposition 5.1). The ratio of the flow parameter to $\tau_2\Delta$ then increases with s to $5/2$ and exceeds 2 exactly when $s \ge 38$; the first Desarguesian example is $\mathrm{PG}(2,41)$, whose cone has 3447 vertices.

Three things survive the counterexamples. The cut parameter satisfies $\tau_c \le \tau_2$ [1, Corollary 4.37], with no factor $\log n$ or Δ at all; the cones show that the flow parameter can be far from it, since on the cones over Ramanujan graphs $\tau_c \le \tau_2 \le (d+1)/(d+1-2\sqrt{d-1})$ while the flow parameter tends to $2(d+1)$. The diameter form of (2) holds on graphs of bounded degree, with a constant of order the logarithm of the maximum degree (Proposition 6.1), and the cones show that this dependence on the degree is nearly sharp. And it holds with $K=2$ on edge-transitive graphs. Whether it holds on regular graphs, or on vertex-transitive graphs, is not settled here (Section 6).

Throughout, $G=(V,E)$ is a finite connected simple graph with $m=|E|$ edges and the chain is the simple random walk, so that $\pi_x = \deg(x)/2m$ and $\pi_x p_{xy} = 1/2m$ on every edge; since $|f_{ij}| = |f_{ji}|$ the maximum in (1) may be taken over undirected edges. We write

$$F(G) = \inf_{\mathbf{f}} \psi(\mathbf{f}) = 2m \inf_{\mathbf{f}} \max_{e\in E} \sum_{a\in V} \pi_a |f_e^{(a)}|. \tag{4}$$

Since simple random walks are reversible chains, a counterexample in this class answers (ii) as it is posed; for general chains Remark 2.2 gives a much simpler one, and the point of the results below is that transitions of small probability are not needed. In continuous time, $q_{ij} = p_{ij}$ for the simple random walk, ψ and τ_2 are unchanged, and so are all statements below.

2. A lower bound from edge lengths

For $\ell : E \to [0,\infty)$ let d_ℓ be the shortest-path metric of the edge lengths ℓ.

Lemma 2.1. *For every $\ell : E \to [0,\infty)$ with $\sum_e \ell_e > 0$,*

$$F(G) \;\ge\; \frac{2m \sum_{a,v\in V} \pi_a \pi_v \, d_\ell(a,v)}{\sum_{e\in E} \ell_e}.$$

Proof. Fix a and a unit flow $\mathbf{f}$ from a to π, and put $\varphi(v) = d_\ell(a,v)$, so that $\varphi(a)=0$ and $|\varphi(i)-\varphi(j)| \le \ell_{ij}$ on every edge. Using $f_{(v)} = 1_{(v=a)} - \pi_v$ and antisymmetry,

$$\begin{aligned}\sum_v \pi_v \varphi(v) &= -\sum_v f_{(v)}\varphi(v) = -\sum_{v,j} f_{vj}\varphi(v) = \frac12 \sum_{v,j} f_{vj}\big(\varphi(j)-\varphi(v)\big) \\ &\le \frac12 \sum_{(v,j)} |f_{vj}|\,\ell_{vj} = \sum_{e\in E} |f_e|\,\ell_e.\end{aligned}$$

Multiply by π_a, sum over a, and use $\sum_a \pi_a |f_e^{(a)}| \le \psi(\mathbf{f})\,\pi_i p_{ij} = \psi(\mathbf{f})/2m$ on each edge:

$$\sum_{a,v} \pi_a \pi_v d_\ell(a,v) \le \sum_e \ell_e \sum_a \pi_a |f_e^{(a)}| \le \frac{\psi(\mathbf{f})}{2m} \sum_e \ell_e.$$

Take the infimum over $\mathbf{f}$. $\square$

Lemma 2.1 is the weak-duality half of linear-programming duality for the maximum concurrent flow problem with demand $\pi_a\pi_v$ between each ordered pair and capacity $1/2m$ on each edge [14, 9]; strong duality, which is used below only for edge-transitive graphs (Section 6), says that the bound is attained for some ℓ, and the linear programs of Section 7 compute that optimum. Taking ℓ to be the indicator of the edges leaving a set A gives $F(G) \ge 2\pi(A)\pi(A^c)/Q(A,A^c)$, so $F(G) \ge 2\tau_c$. Conversely, for this ℓ let $C_1,\dots,C_k$ be the components of the graph obtained from G by deleting the edges leaving A. Then $d_\ell(a,v) = 0$ for a, v in the same component and $d_\ell(a,v) \le \Delta$ otherwise. Every edge leaving some C_i leaves A, and every edge leaving A leaves exactly two of the C_i, so $\sum_i Q(C_i, C_i^c) = 2Q(A,A^c)$; since $\pi(C_i)\pi(C_i^c) \le \tau_c Q(C_i, C_i^c)$ for each i, the certificate is at most

$$\frac{2\Delta \sum_i \pi(C_i)\pi(C_i^c)}{\sum_i Q(C_i, C_i^c)} \;\le\; 2\tau_c\Delta \;\le\; 2\tau_2\Delta.$$

So a cut certificate never certifies a ratio $F/(\tau_2\Delta)$ above 2. The value 2 is attained by the complete graph, for which $F = 2(n-1)/n$ (Lemma 2.1 with $\ell \equiv 1$, and the flows that send π_v from a directly to each v), $\tau_2 = (n-1)/n$ and $\Delta = 1$; constants below 2 therefore fail for trivial reasons, and among simple random walks a counterexample with $K \ge 2$ must have $\Delta \ge 2$. The certificates used below are not cut certificates.

Remark 2.2 (general chains)**.** For general reversible chains the answer to (ii) is negative by a device that Aldous and Fill use for the n-cycle in the paragraph after Corollary 4.22 of [1]. The proof of Lemma 2.1 applies to any reversible chain, with $\pi_i p_{ij}$ in place of $1/2m$, and gives

$$\inf_{\mathbf{f}} \psi(\mathbf{f}) \;\ge\; \frac{\sum_{a,v} \pi_a\pi_v\, d_\ell(a,v)}{\sum_{\{i,j\}} \ell_{ij}\, \pi_i p_{ij}}$$

for every length function ℓ on the edges of the transition graph. Let H be a connected d-regular graph on n vertices, P the transition matrix of the simple random walk on H, J the all-ones matrix, $0 < \varepsilon \le 1/(n^2 \operatorname{diam} H)$, and $P_\varepsilon = (1-(n-1)\varepsilon)P + \varepsilon(J-I)$. The matrix P_ε is symmetric, so its stationary distribution is uniform, and its transition graph is complete, so $\Delta = 1$. Its eigenvalues other than 1 are $(1-(n-1)\varepsilon)\lambda - \varepsilon$ for the eigenvalues $\lambda \ne 1$ of P, and $\lambda_2(P) \ge -1/(n-1)$ because the trace of P is 0, so $\tau_2(P_\varepsilon) \le \tau_2(P)$. Take $\ell = 1$ on the edges of H and $\ell = \operatorname{diam} H$ on the other pairs. Then $d_\ell = \operatorname{dist}_H$, and the denominator equals $\frac12 + \frac12(n-1-d)\varepsilon(\operatorname{diam} H - 1) \le \frac12(1+\frac1n)$, so

$$\inf_{\mathbf{f}} \psi(\mathbf{f}) \;\ge\; \frac{2}{1+1/n} \cdot \frac{1}{n^2} \sum_{a,v} \operatorname{dist}_H(a,v) \;\ge\; (2-o(1)) \log_{d-1} n$$

for fixed $d \ge 3$, since at most $1 + B_r$ vertices of H lie within distance $r-1$ of a given vertex, with B_r as in Theorem 1.2. If H is a Ramanujan graph then $\tau_2(P_\varepsilon) \le d/(d-2\sqrt{d-1})$, and the ratio $\inf_{\mathbf{f}} \psi(\mathbf{f})/(\tau_2\Delta)$ is of order $\log n$.

3. Cones over regular graphs

Let H be a connected d-regular graph on n vertices with $d \ge 3$ and $n \ge d+2$, so that H is not complete, and let $G = K_1 \vee H$ be the cone over H: $V(G) = V(H) \cup \{h\}$ with the hub h adjacent to every vertex of H. Then G has $n+1$ vertices,

$$m = \frac{nd}{2} + n = \frac{n(d+2)}{2}, \qquad \deg h = n, \qquad \deg v = d+1 \quad (v \in V(H)),$$

$$\pi_h = \frac{1}{d+2}, \qquad \pi_v = \frac{d+1}{n(d+2)}, \qquad \pi_B := \pi(V(H)) = \frac{d+1}{d+2}.$$

Let $d = \mu_1 \ge \mu_2 \ge \cdots \ge \mu_n$ be the adjacency eigenvalues of H.

Lemma 3.1. $\Delta(G) = 2$. *The eigenvalues of the transition matrix P of the simple random walk on G are 1, $-1/(d+1)$ and $\mu_i/(d+1)$ for $i = 2, \dots, n$; hence $\lambda_2(P) = \mu_2/(d+1)$ and $\tau_2(G) = (d+1)/(d+1-\mu_2)$.*

Proof. Every vertex is adjacent to h, so any two vertices are at distance at most 2, and since H is not complete some two base vertices are not adjacent; so $\Delta = 2$. Let $z \perp \mathbf{1}$ be an eigenvector of the adjacency matrix of H with eigenvalue μ, extended by $z_h = 0$. At a base vertex v, $(Pz)_v = \frac{1}{d+1}\big(\sum_{w\sim v,\, w\in V(H)} z_w + z_h\big) = \frac{\mu}{d+1} z_v$, and $(Pz)_h = \frac{1}{n}\sum_v z_v = 0$. This gives $n-1$ independent eigenvectors with eigenvalues $\mu_i/(d+1)$, $i \ge 2$. On the complementary space of vectors constant on $V(H)$, in the basis (value at h, value on the base), P acts by $\left(\begin{smallmatrix} 0 & 1 \\ 1/(d+1) & d/(d+1) \end{smallmatrix}\right)$, whose eigenvalues are 1 and $-1/(d+1)$. Finally, H contains two non-adjacent vertices, and Cauchy interlacing with the induced subgraph on them gives $\mu_2 \ge 0$, so the second largest eigenvalue of P is $\mu_2/(d+1)$. □

Lemma 3.2. $F(G) \le 2(d+1)(1-\pi_v) < 2(d+1)$.

Proof. Route everything through the hub. For a base vertex a let $f^{(a)}_{ah} = 1 - \pi_a$ and $f^{(a)}_{hw} = \pi_w$ for every base vertex $w \ne a$; for the hub let $f^{(h)}_{hw} = \pi_w$ for every base w; all other values are zero or forced by antisymmetry. Each $\mathbf{f}^{(a)}$ is a unit flow from a to π: for $a \ne h$ the net outflow at h is $(\pi_B - \pi_a) - (1-\pi_a) = -\pi_h$, at a it is $1-\pi_a$, and at every other base vertex w it is $-\pi_w$. Base edges carry nothing. On the spoke $\{h, v\}$ the load is

$$\sum_a \pi_a |f^{(a)}_{hv}| = \pi_v(1-\pi_v) + \pi_h\pi_v + \sum_{w\in V(H),\, w\ne v} \pi_w\pi_v = 2\pi_v(1-\pi_v),$$

so $\psi(\mathbf{f}) = 2m \cdot 2\pi_v(1-\pi_v) = 2(d+1)(1-\pi_v)$, since $2m\pi_v = \deg v = d+1$. □

Lemma 3.3. *For an integer $r \ge 1$ let $B_r = d + d(d-1) + \cdots + d(d-1)^{r-2} = d\big((d-1)^{r-1} - 1\big)/(d-2)$, with $B_1 = 0$, and $L = (d+2)r/(r+d)$. Then*

(5)

$$F(G) \;\ge\; V_r := 2L\Big[\pi_h\pi_B + \pi_v^2 \sum_{u\ne w\in V(H)} \min\Big(\frac{\mathrm{dist}_H(u,w)}{r}, 1\Big)\Big] \;\ge\; 2(d+1)\,\frac{r}{r+d}\Big(1 - \frac{1+B_r}{n}\Big).$$

Proof. Give every base edge the length $b = 2L/r = 2(d+2)/(r+d)$ and every spoke the length L. Then $\sum_e \ell_e = \frac{nd}{2} b + nL = nL\big(\frac{d}{r} + 1\big) = n(d+2) = 2m$, so Lemma 2.1 reads $F(G) \ge \sum_{a,v} \pi_a\pi_v d_\ell(a,v)$. A path from h to a base vertex begins with a spoke, so $d_\ell(h,v) = L$. A path between base vertices $u \ne w$ either stays in H, with length at least $b\,\mathrm{dist}_H(u,w)$, or passes through h, with length at least $2L$, and both are attained; so $d_\ell(u,w) = \min(b\,\mathrm{dist}_H(u,w), 2L) = 2L\min(\mathrm{dist}_H(u,w)/r, 1)$. Summing over ordered pairs gives V_r.

In a graph of maximum degree d at most B_r vertices lie at distance between 1 and $r-1$ from a given vertex, so at least $n(n-1) - nB_r$ ordered pairs of distinct base vertices are at distance at least r in H. Hence

$$V_r \ge 2L\big[\pi_h\pi_B + \pi_v^2\, n(n-1-B_r)\big] = 2L\pi_B\Big[\pi_h + \pi_B - \pi_B\frac{1+B_r}{n}\Big] \ge 2L\pi_B\Big(1 - \frac{1+B_r}{n}\Big),$$

and $L\pi_B = (d+1)r/(r+d)$. □

Proof of Theorem 1.2. Lemmas 3.1 to 3.3 give the first two statements and (3). In (3) let $r = r_n \to \infty$ so slowly that $(d-1)^{r_n} = o(n)$; then $B_{r_n}/n \to 0$ and $r_n/(r_n+d) \to 1$, so $F(G) \to 2(d+1)$, which uses only the two-sided bound and holds for every sequence. If moreover $\mu_2(H) \le \beta$ then $\tau_2\Delta \le 2(d+1)/(d+1-\beta)$ by Lemma 3.1, and the ratio has liminf at least $d+1-\beta$. □

Remark 3.4. The mechanism is visible in the numbers. The spokes carry total stationary flow rate $n/2m = \pi_h = 1/(d+2)$, and all traffic to or from a base vertex must leave it along its single spoke or along one of its d base edges. The base has $d/2$ times the capacity of the spokes, but in a d-regular graph on $n \gg (d-1)^r$ vertices almost all pairs are at distance at least r, so routing in the base costs at least r units of capacity per unit of demand against 2 through the hub; once $r \gg d$ the base is useless for typical pairs, and each spoke is a bottleneck with congestion $2(d+1)$. The cut bound sees none of this: the cut separating h from $V(H)$ gives only $2\pi_h\pi_B/\pi_h = 2(d+1)/(d+2)$.

4. Proof of Theorem 1.1

Let $p \equiv 1 \pmod 4$ be prime. For every prime $q \equiv 1 \pmod 4$ with $q \neq p$ and $q > 2\sqrt{p}$, Lubotzky, Phillips and Sarnak [10] construct a connected $(p+1)$-regular Cayley graph $X^{p,q}$, of $\mathrm{PSL}_2(\mathbb{F}_q)$ when p is a quadratic residue modulo q (then $X^{p,q}$ is not bipartite and has $q(q^2-1)/2$ vertices) and of $\mathrm{PGL}_2(\mathbb{F}_q)$ otherwise (then it is bipartite with $q(q^2-1)$ vertices), whose eigenvalues other than $\pm(p+1)$ all satisfy $|\mu| \le 2\sqrt{p}$. By Dirichlet's theorem there are infinitely many primes $q \equiv 1 \pmod{4p}$, and for these p is a residue modulo q by quadratic reciprocity. Apply Theorem 1.2 to the non-bipartite graphs $X^{p,q}$ with $q \to \infty$, $d = p+1$ and $\beta = 2\sqrt{p}$:

$$\liminf_{q\to\infty} \frac{F(K_1 \vee X^{p,q})}{\tau_2 \Delta} \;\ge\; p + 2 - 2\sqrt{p} = (\sqrt{p}-1)^2 + 1 \xrightarrow[p\to\infty]{} \infty.$$

Given K, choose p with $(\sqrt{p}-1)^2 + 1 > K$ and then q large. Morgenstern's $(p+1)$-regular Ramanujan graphs [12], which exist for every prime power p, give the same with $d = p+1$, and Friedman's theorem [6], in the form for the uniform model given by Bordenave [2, Theorem 1], by which a uniformly random d-regular graph on n vertices has all eigenvalues other than d of absolute value at most $2\sqrt{d-1}+\varepsilon$ with probability tending to one, gives it for every $d \ge 3$ with $\beta = 2\sqrt{d-1}+\varepsilon$. □

The quantifiers can be made explicit. With $r = 3d$ and $n \ge 2(1+B_{3d})$, (3) gives $F/(\tau_2\Delta) \ge \frac{3}{8}(d+1-\mu_2)$; so for $p = 101$, $d = 102$, any Ramanujan base on $n \ge 2(1+B_{306})$ vertices gives a ratio above $\frac{3}{8}(103 - 2\sqrt{101}) > 30$.

Remark 4.1 (absolute spectral gap)**.** With the non-bipartite bases of the proof, every eigenvalue of P other than 1 has absolute value at most $2\sqrt{d-1}/(d+1)$ by Lemma 3.1, so the conclusion also holds with τ_2 replaced by the absolute relaxation time $(1 - \max(\lambda_2, |\lambda_n|))^{-1}$. This is why the bipartite members of the family are avoided: a bipartite base contributes the eigenvalue $-d/(d+1)$, and the absolute relaxation time would then be $d+1$.

Remark 4.2 (the length-weighted parameter)**.** For a family presented as path flows, with $\mathbf{f}^{(a)}$ decomposed into simple $a \to v$ paths γ carrying total mass π_v, Sinclair [15], modifying a bound of Diaconis and Stroock [4] for canonical paths, uses

$$\bar\rho(\mathbf{f}) = \max_{(i,j)} \frac{1}{\pi_i p_{ij}} \sum_a \pi_a \sum_{\gamma \ni (i,j)} |\gamma|\, f^{(a)}(\gamma),$$

where the inner sum is over the paths that traverse (i,j) from i to j and $|\gamma|$ is the number of edges of γ. The value $|f_{ij}^{(a)}|$ is at most the mass of the paths of $\mathbf{f}^{(a)}$ through (i,j) plus the mass of those through (j,i); since $|\gamma| \ge 1$ and $\pi_i p_{ij} = \pi_j p_{ji}$, every path family has $\psi(\mathbf{f}) \le 2\bar\rho(\mathbf{f})$, so $\inf \bar\rho \ge F(G)/2$. The hub routing of Lemma 3.2 uses paths of length at most 2, and each of its flows crosses every edge in one direction only, so it has $\bar\rho \le 2\psi < 4(d+1)$. On the cones $\inf \bar\rho$ therefore lies between $F(G)/2$ and $4(d+1)$, and $\inf \bar\rho/(\tau_2\Delta)$ is unbounded as well: the answer to (ii) is negative in either normalisation. In this normalisation the cones are not needed.

Since the weights $\pi_i p_{ij}$ of the directed edges add up to at most 1 and a path γ traverses $|\gamma|$ directed edges, multiplying the definition of $\bar\rho$ by $\pi_i p_{ij}$ and summing over the directed edges (i,j) gives $\bar\rho(\mathbf{f}) \geq \sum_a \pi_a \sum_\gamma |\gamma|^2 f^{(a)}(\gamma) \geq \sum_{a,v} \pi_a \pi_v \operatorname{dist}(a,v)^2$. On d-regular Ramanujan graphs of fixed degree this is at least of order $(\log n)^2$, while τ_2 is bounded and Δ is of order $\log n$; so $\inf \bar\rho/(\tau_2\Delta)$ is unbounded already on graphs of bounded degree, where $\inf_{\mathbf{f}} \psi(\mathbf{f})/(\tau_2\Delta)$ is bounded by Proposition 6.1. Kahale [8] studies bounds of this kind with variable lengths on the edges; from multicommodity flows he obtains a bound on the spectral gap that can be computed by semidefinite programming and is within a factor $O(\log^2 n)$ of the spectral gap for every reversible chain on n states, and this factor cannot be improved in general.

5. Projective-plane cones

A projective plane Π of order $s \geq 2$ has $N = s^2+s+1$ points and N lines; every line contains $s+1$ points, every point lies on $s+1$ lines, any two points lie on exactly one common line, and any two lines meet in exactly one point. For every prime power s the Desarguesian plane $\mathrm{PG}(2,s)$ is an example. The incidence graph $\mathrm{Inc}(\Pi)$ is the bipartite graph on the $n = 2N$ points and lines in which a point is adjacent to the lines through it. It is $(s+1)$-regular, and it has no cycle of length four, since two points lie on only one common line.

Proposition 5.1. *Let Π be a projective plane of order $s \geq 2$, $H = \mathrm{Inc}(\Pi)$ and $G = K_1 \vee H$. Then $F(G)$ equals the certificate V_3 of Lemma 3.3, and*

$$F(G) = \frac{(s+2)(5s^3+19s^2+13s+8)}{(s+3)(s+4)(s^2+s+1)}, \qquad \tau_2(G)\,\Delta(G) = \frac{2(s+2)}{s+2-\sqrt{s}}.$$

As a function of s, the ratio $F(G)/(\tau_2\Delta)$ is strictly increasing on $[2,\infty)$ and tends to $5/2$, and for integer $s \geq 2$ it exceeds 2 if and only if $s \geq 38$.

Proof. Let M be the point-line incidence matrix. Since two distinct points lie on exactly one common line, $MM^{\mathsf T} = sI + J$, with eigenvalues $(s+1)^2$, once, and s, with multiplicity $N-1$. The adjacency eigenvalues of H are plus and minus the singular values of M, that is $\pm(s+1)$ and $\pm\sqrt{s}$, so $\mu_2(H) = \sqrt{s}$, and Lemma 3.1 with $d = s+1$ gives $\tau_2\Delta = 2(s+2)/(s+2-\sqrt{s})$.

Next, the distances in H. From a point p, the $s+1$ lines through p are at distance 1; every other point q is at distance 2, and the line through p and q is their only common neighbour; and each of the s^2 lines ℓ not through p is at distance 3 and is joined to p by exactly $s+1$ paths of length 3, namely $p, m, m\cap\ell, \ell$ for the lines m through p. By duality the same holds from a line. So every base vertex has $s+1$, s^2+s and s^2 base vertices at distance 1, 2 and 3.

Lower bound. In (5) with $r = 3$ the sum over pairs of base vertices equals $n\big(\frac{s+1}{3} + \frac{2(s^2+s)}{3} + s^2\big) = \frac{n}{3}(5s^2+3s+1)$. With $L = 3(s+3)/(s+4)$, $\pi_h = 1/(s+3)$, $\pi_v = (s+2)/(n(s+3))$ and $\pi_B = (s+2)/(s+3)$ this gives

$$V_3 = \frac{6(s+3)}{s+4}\cdot\frac{s+2}{(s+3)^2}\Big(1 + \frac{(s+2)(5s^2+3s+1)}{6N}\Big) = \frac{(s+2)(5s^3+19s^2+13s+8)}{(s+3)(s+4)N},$$

because $6N + (s+2)(5s^2+3s+1) = 5s^3+19s^2+13s+8$.

Upper bound. Let

$$\theta = \frac{3(s^2+3s+1)}{s(s+2)(s+4)}, \qquad \text{so that} \qquad 1-\theta = \frac{(s-1)(s+1)(s+3)}{s(s+2)(s+4)} \quad \text{and} \quad 0 < \theta \leq 1.$$

The hub sends π_v to each base vertex v along the spoke hv. A base vertex a sends π_h to the hub along its spoke, sends π_v to each v at distance 1 or 2 along the shortest path, and sends to each v at distance 3 the amount $\theta\pi_v$ along a, h, v and the amount $(1-\theta)\pi_v/(s+1)$ along each of the $s+1$ paths of length 3. Each of these is a superposition of path flows that carry the

mass π_x from the source to every other vertex x, so it is a unit flow from the source to π. In the flow from a base vertex a no edge is used in both directions: the paths in H are shortest paths, along which the distance from a increases at each step, the spoke at a is used towards h, and the other spokes are used towards the base. The flow from h uses only spokes, each towards the base. So the load $\sum_x \pi_x |f_e^{(x)}|$ of an edge e is the sum, over the paths through e, of the mass of the path times π_x, where x is its source.

The spoke hv is used by the flow from h, with weight $\pi_h\pi_v$; by the flow from v, with weight $\pi_v(\pi_h + \theta s^2\pi_v)$; and by the flows from the s^2 base vertices at distance 3 from v, with weight $\theta\pi_v^2$ each. Its load is therefore $2\pi_h\pi_v + 2\theta s^2\pi_v^2$. A base edge uw is used by the pairs (u,w) and (w,u); by the $4s$ ordered pairs at distance 2 whose path begins or ends with uw, the other end being one of the s further neighbours of u or of w; and by $6s^2$ paths of length 3, since it can be the first, second or third edge of such a path, in either direction, with s choices for each of the two other edges, and every path of length 3 in H joins two vertices at distance 3 because H is bipartite and has no cycle of length four. Its load is therefore

$$\pi_v^2\Big(2 + 4s + \frac{6s^2(1-\theta)}{s+1}\Big).$$

Since $\pi_h/\pi_v = n/(s+2) = 2N/(s+2)$, the two loads are equal if and only if

$$\frac{2s^2(s+4)}{s+1}\,\theta = \frac{(10s^2+6s+2)(s+2) - 4N(s+1)}{(s+1)(s+2)} = \frac{6s(s^2+3s+1)}{(s+1)(s+2)},$$

which is the definition of θ. So every edge carries the same load, and since $2m = n(s+3)$, $2m\pi_v = s+2$ and $2N(s+4) + 3s(s^2+3s+1) = 5s^3 + 19s^2 + 13s + 8$,

$$\psi(\mathbf{f}) = 2m\big(2\pi_h\pi_v + 2\theta s^2\pi_v^2\big) = \frac{s+2}{s+3}\Big(2 + \frac{3s(s^2+3s+1)}{N(s+4)}\Big) = V_3.$$

With the lower bound, $F(G) = V_3$.

The ratio. We have $F/(\tau_2\Delta) = \frac{1}{2}V_3\big(1 - \frac{\sqrt{s}}{s+2}\big)$. The derivative of V_3 in s is

$$\frac{11s^6 + 98s^5 + 355s^4 + 734s^3 + 949s^2 + 584s + 104}{(s+3)^2(s+4)^2(s^2+s+1)^2} > 0,$$

and $\sqrt{s}/(s+2)$ is strictly decreasing on $[2,\infty)$: it equals $a/(a^2+2)$ with $a = \sqrt{s}$, and

$$\frac{a}{a^2+2} - \frac{b}{b^2+2} = \frac{(b-a)(ab-2)}{(a^2+2)(b^2+2)} > 0 \qquad \text{for } \sqrt{2} \le a < b.$$

Both factors are positive, so the ratio is strictly increasing on $[2,\infty)$, and it tends to $5/2$ because $V_3 \to 5$. The ratio exceeds 2 exactly when $(V_3 - 4)(s+2) > V_3\sqrt{s}$. At $s = 37$ and $s = 38$,

$$V_3(37) = \frac{727389}{153832}, \qquad V_3(38) = \frac{2015320}{425621},$$

and $(V_3 - 4)(s+2)/V_3$ equals $112061/18651 < 6.01 < \sqrt{37}$ at $s = 37$ and $312836/50383 > 6.2 > \sqrt{38}$ at $s = 38$. □

Projective planes exist for every prime-power order, and 41 is the least prime power above 37, so the first Desarguesian cone on which the ratio exceeds 2 is $K_1 \vee \mathrm{Inc}(\mathrm{PG}(2,41))$: it has 3447 vertices, $F = 1080977/227436$ and ratio 2.0226. No plane of order 38 exists, by the Bruck–Ryser theorem [3], and none of order 39 or 40 is known, so 41 is also the smallest order of a known projective plane whose cone has ratio above 2. The degree $s+1$ of these bases grows with the number of vertices, so the limit in Theorem 1.2 does not apply to them, and their ratio stays

below 5/2. The cones over Ramanujan graphs of a fixed degree $d \geq 6$ exceed 5/2 in the limit, since $d+1-2\sqrt{d-1} \geq 7-2\sqrt{5} > 5/2$.

Other generalized polygons [16] give larger ratios in the limit, though not at comparable sizes. The incidence graph of a generalized m-gon of order (s,s) is $(s+1)$-regular with $2(s^m-1)/(s-1)$ vertices, girth $2m$ and diameter m, so every vertex has $(s+1)s^{i-1}$ vertices at distance i for $1 \leq i < m$ and s^{m-1} at distance m; and $\mu_2 = 2\sqrt{s}\cos(\pi/m)$ [5]. For $m=3$ these are the projective planes. On the cone, the certificate V_m of Lemma 3.3 gives $V_m/(\tau_2\Delta) \to m - \frac{1}{2}$ as $s \to \infty$; rounded down, the bound is 2.035 on the cone over the symplectic quadrangle $W(16)$, which has 8739 vertices, and 2.193 on the cone over the split Cayley hexagon $H(11)$, which has 354,313 vertices, while by Proposition 5.1 the cones over $\mathrm{PG}(2,64)$ and $\mathrm{PG}(2,419)$, with 8323 and 351,963 vertices, give 2.125 and 2.366. Since a generalized m-gon of order (s,s) with $m \geq 3$ and $s \geq 2$ exists only for $m = 3, 4, 6$ [5], the limits of these certificates are at most 11/2.

6. What remains

Proposition 6.1. *For every finite reversible discrete-time chain,*

$$\inf_{\mathbf{f}} \psi(\mathbf{f}) \;\leq\; \frac{16e}{e-1}\,\tau_2\Big(1+\frac{1}{2}\log\frac{1}{\pi_*}\Big), \qquad \pi_* = \min_i \pi_i.$$

For the simple random walk on a connected simple graph with maximum degree $D \geq 3$ and diameter Δ,

$$\inf_{\mathbf{f}} \psi(\mathbf{f}) \;\leq\; 65\,\tau_2\,\Delta\log D.$$

Proof. For uniform π the first inequality is the remark in the notes to Section 4.3.5 of [1] that (2) follows from Corollary 4.22 and Lemma 4.23 there; we write it out for general π in discrete time. By Lemma 4.21 of [1], $\inf_{\mathbf{f}}\psi(\mathbf{f}) \leq \psi(\mathbf{f}^{a\to\pi}) \leq \tau_1^{(4)}$; by Lemmas 4.13 and 4.12 there, $\tau_1^{(4)} \leq 4\tau_1^{(3)} \leq 4\tau_1^{(2)}$; and by Lemma 4.23 there, $\tau_1^{(2)} \leq \frac{4e}{e-1}\tau_2(1+\frac{1}{2}\log\frac{1}{\pi_*})$ in discrete time. For a graph of maximum degree $D \geq 3$ and diameter Δ, $n \leq 1 + D\big((D-1)^{\Delta}-1\big)/(D-2) \leq 4D^{\Delta}$ and, with $\delta_{\min}$ the least degree, $1/\pi_* = 2m/\delta_{\min} \leq nD \leq 4D^{\Delta+1}$, so $1+\frac{1}{2}\log\frac{1}{\pi_*} \leq 1+\log 2+\frac{1}{2}(\Delta+1)\log D \leq 2.55\,\Delta\log D$, and $\frac{16e}{e-1}\cdot 2.55 < 65$. □

So the diameter form of (2) holds with a constant depending on the maximum degree, $K(D) = O(\log D)$, and in particular on every class of graphs of bounded degree. The cones show that the dependence on D is nearly sharp. For every $d \geq 3$, repeated 2-lifts of $K_{d,d}$ give bipartite d-regular graphs on $2d\cdot 2^k$ vertices, $k \geq 0$, whose eigenvalues other than $\pm d$ lie in $[-2\sqrt{d-1}, 2\sqrt{d-1}]$ [11]; these graphs are connected, because d is a simple eigenvalue of $K_{d,d}$ and each lift adds only eigenvalues in $[-2\sqrt{d-1}, 2\sqrt{d-1}]$, and they have $\mu_2 \leq 2\sqrt{d-1}$, and one of them, H, has $2(1+B_{3d}) \leq n < 4(1+B_{3d})$. The cone $K_1 \vee H$ has $D = n$, the choice $r = 3d$ in (3) gives the first inequality below as in the paragraph after the proof of Theorem 1.1, and $1 + B_{3d} \leq d^{3d}$ gives $\log n \leq 3d\log d + \log 4$, so $d \geq c\log n/\log\log n$ and

$$\frac{F(K_1\vee H)}{\tau_2\Delta} \;\geq\; \frac{3}{8}\big(d+1-2\sqrt{d-1}\big) \;\geq\; c'\,\frac{\log D}{\log\log D}$$

with absolute constants $c, c' > 0$. Closing the gap between $\log D/\log\log D$ and $\log D$ is open.

Question 6.2. Does the diameter form of (2) hold with a universal constant for simple random walks on regular graphs, or on vertex-transitive graphs?

The mechanism of Section 3 needs one vertex of degree comparable to n next to vertices of bounded degree. In a regular graph of diameter two every degree is at least $\sqrt{n-1}$, so no vertex

can send most of its edges into a sparse part of the graph, and we do not know whether some other mechanism makes the ratio unbounded on regular graphs.

On edge-transitive graphs the diameter form holds with $K = 2$. By strong duality $F(G)$ is the maximum over ℓ of the right side of Lemma 2.1. With $\sum_e \ell_e = 1$ the right side is a concave function of ℓ, since $d_\ell(a, v)$ is a minimum of linear functions, and it is invariant under the automorphisms of G; averaging an optimal ℓ over the automorphism group therefore shows that a constant ℓ is optimal when the group acts transitively on the edges. Then $F(G) = 2\sum_{a,v} \pi_a \pi_v \operatorname{dist}(a, v) < 2\Delta$. If G is not complete then $\lambda_2 \geq 0$, since P is similar to the symmetric matrix with entries $a_{ij}/\sqrt{\deg(i)\deg(j)}$, where (a_{ij}) is the adjacency matrix, and its principal submatrix on two non-adjacent vertices is zero; Cauchy interlacing applies as in the proof of Lemma 3.1. Hence $\tau_2 \geq 1$ and $F(G) < 2\tau_2\Delta$, while the complete graph has $F = 2\tau_2\Delta$. The vertex-transitive case of the question therefore concerns graphs with more than one orbit on edges.

7. Computations

$F(G)$ is the optimal value of the linear program

$$\min \lambda \quad \text{subject to} \quad \sum_j \bigl(x^a_{ij} - x^a_{ji}\bigr) = 1_{(i=a)} - \pi_i \quad (a, i \in V),$$

$$\sum_a \deg(a)\bigl(x^a_{e^+} + x^a_{e^-}\bigr) \leq \lambda \quad (e \in E), \qquad x \geq 0,$$

where e^+ and e^- are the two orientations of e; this is (4) after the substitution $f^{(a)}_e = x^a_{e^+} - x^a_{e^-}$ and multiplication of the congestion constraint by $2m$. It was solved with the HiGHS solver [7] through SciPy [17], so the values carry the solver's tolerance, about 10^{-8}. Every certificate of Lemma 2.1 was evaluated in exact rational arithmetic, with Dijkstra's algorithm on integer weights for general length functions, so each certificate is a rigorous lower bound on F for the graph in question; the tables print it rounded to four decimals. The linear program reproduces $F(K_n) = 2(n-1)/n$ for $3 \leq n \leq 6$, $F(K_1 \vee C_5) = 21/10$, and $F = 64/25$ and $133/52$ for the cones over the Petersen graph and the Paley graph on 13 vertices; the eigenvalues of P on cones over random regular graphs, over the Petersen graph and over the bipartite incidence graph of $\mathrm{PG}(2, 3)$ agree with Lemma 3.1 to 10^{-15}; and the hub routing, built as an explicit family and evaluated exactly, has $\psi = 2(d+1)(1-\pi_v)$.

On the projective-plane cones the linear program agrees with Proposition 5.1 to the solver's tolerance (Table 1), and the routing in its proof, built as an explicit family and evaluated in exact arithmetic for $s = 2, 3, 4, 5$, has $2m\sum_x \pi_x |f^{(x)}_e| = V_3$ on every edge e. The random regular graphs of Tables 2 and 3 are those generated by `networkx.random_regular_graph` in NetworkX 3.6.1 with seed 1, which gives a connected graph for every size used. On cones over random regular graphs with up to 160 base vertices (Table 2) F and the best certificate rise slowly with n towards $2(d+1)$, but these sizes are far from the regime of Theorem 1.2: the certificate needs most pairs of H at distance at least r with r several times d, hence $n \gg (d-1)^{3d}$, which is about $10^{2.7}$ for $d = 3$, $10^{5.7}$ for $d = 4$ and 10^9 for $d = 5$, while the linear program can be solved up to a few hundred vertices and the exact certificate up to some 10^4 vertices. Table 3 shows the certificates on larger random cones and on the cones over the Lubotzky–Phillips–Sarnak graphs $X^{5,q}$, built from the six quaternions of norm 5 as 2×2 matrices over $\mathbb{F}_q$; the computed μ_2 confirms the Ramanujan bound $2\sqrt{5}$. In Table 2 the linear program stays well below the hub bound, so routing in the base still competes with the hub, and the ratio lies between 0.66 and 0.87; in Table 3 the certificates bound the ratio below by 0.798 to 1.200. The regime in which the ratio exceeds every constant is reached by the proof, not by computation.

TABLE 1. Cones over the incidence graphs of $\mathrm{PG}(2,s)$: the linear program against the closed form of Proposition 5.1, the relaxation time $\tau_2 = (s+2)/(s+2-\sqrt{s})$, and the ratio $F/(\tau_2\Delta)$ with $\Delta = 2$. For $s \geq 37$ only the closed form is shown.

s	$\lvert V(G)\rvert$	F (linear program)	F (Proposition 5.1)	τ_2	$F/(\tau_2\Delta)$
2	15	2.857142857	20/7	1.546918	0.9235
3	27	3.232600733	1765/546	1.530012	1.0564
4	43	3.489795918	171/49	1.500000	1.1633
5	63	3.678763441	2737/744	1.469374	1.2518
7	115	3.940191388	1647/418	1.416375	1.3909
37	2815	—	727389/153832	1.184790	1.9955
41	3447	—	1080977/227436	1.174964	2.0226
49	4903	—	10786653/2251652	51/44	2.0665

TABLE 2. Cones over random d-regular graphs on n vertices. Columns: $\mu_2(H)$; τ_2; the diameter of H; the best certificate $\max_r V_r$ of (5) (computed exactly, rounded to four decimals) and its r; F by the linear program; the hub bound of Lemma 3.2; the ratio $F/(\tau_2\Delta)$; and the limiting value $d+1-\mu_2$ that the ratio would take if F were $2(d+1)$.

d	n	$\mu_2(H)$	τ_2	diam H	$\max_r V_r$	r	F	hub bound	$F/(\tau_2\Delta)$	$d+1-\mu_2$
3	20	2.6659	2.9984	6	3.3600	4	4.0000	7.6800	0.667	1.334
3	40	2.7044	3.0875	7	3.8057	4	4.2900	7.8400	0.695	1.296
3	80	2.7383	3.1703	8	4.2873	5	4.4193	7.9200	0.697	1.262
3	160	2.7783	3.2741	9	4.6534	6	4.6978	7.9600	0.717	1.222
4	20	3.0224	2.5284	4	3.1726	3	3.5985	9.5833	0.712	1.978
4	40	3.2585	2.8711	5	3.6443	3	3.8487	9.7917	0.670	1.742
4	80	3.3683	3.0644	6	4.1686	4	4.2047	9.8958	0.686	1.632
5	20	3.1229	2.0854	3	2.9507	3	3.3673	11.4857	0.807	2.877
5	40	3.5751	2.4743	4	3.6048	3	3.6096	11.7429	0.729	2.425
5	80	3.9329	2.9026	5	4.0290	4	4.0943	11.8714	0.705	2.067
6	20	3.5153	2.0088	3	2.8875	2	3.0555	13.3875	0.761	3.485
6	40	4.2180	2.5162	4	3.5284	3	3.6462	13.6937	0.725	2.782
8	40	4.2066	1.8776	3	3.2302	3	3.2432	17.5950	0.864	4.793

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DECLARATION OF COMPETING INTEREST

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TABLE 3. Larger cones, certificates only: cones over random d-regular graphs and over the Lubotzky–Phillips–Sarnak graphs $X^{5,q}$ (6-regular; bipartite with $q(q^2-1)$ vertices for $q = 13, 17$, where 5 is not a square modulo q, and with $q(q^2-1)/2$ vertices for $q = 29$). Columns: $\mu_2(H)$; τ_2; the diameter of H; the best certificate $\max_r V_r$ (computed exactly; before rounding to four decimals it is a rigorous lower bound on F) and its r; the hub bound; the resulting lower bound on $F/(\tau_2\Delta)$, rounded down to three decimals (τ_2 is computed in floating point); and $d+1-\mu_2$.

base	n	d	$\mu_2(H)$	τ_2	diam H	$\max_r V_r$	r	hub bound	$V_r/(\tau_2\Delta)$	$d+1-\mu_2$
random	1000	3	2.8230	3.3985	13	5.4276	8	7.9936	0.798	1.177
random	4000	3	2.8237	3.4006	15	5.8205	10	7.9984	0.855	1.176
random	16000	3	2.8261	3.4073	17	6.1214	11	7.9996	0.898	1.174
random	1000	4	3.4471	3.2199	9	5.4871	6	9.9917	0.852	1.553
random	4000	4	3.4603	3.2474	10	6.0004	7	9.9979	0.923	1.540
$X^{5,13}$	2184	6	4.2497	2.5452	7	5.8695	5	13.9944	1.153	2.750
$X^{5,17}$	4896	6	4.3089	2.6012	9	6.2202	6	13.9975	1.195	2.691
$X^{5,29}$	12180	6	4.4420	2.7365	8	6.5699	6	13.9990	1.200	2.558

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In September 2026 an agent running OpenAI GPT-5.6 selected the question from the open problems of [1] and identified the cone $K_1 \vee \mathrm{Inc}(\mathrm{PG}(2,49))$ as a counterexample to the diameter form of (2) with $K = 2$. The construction of Section 3, the proofs, the computations and the text were then produced with Anthropic's Claude Fable 5.1 in Claude Code. In October 2026 Claude Opus 5.5, also in Claude Code, found the routing in the proof of Proposition 5.1, the argument of Remark 2.2, the bound on cut certificates after Lemma 2.1, the certificates on other generalized polygons in Section 5 and the argument for edge-transitive graphs in Section 6, and revised the text. The proofs were checked by further runs of these systems, and the algebraic identities and numerical inequalities listed in the repository were verified in Lean 4 with Mathlib; the counting arguments and the spectral facts that feed them were not formalized. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

The code that produces the tables (the linear programs, the exact certificates and the construction of the Lubotzky–Phillips–Sarnak graphs), an exact check of the routing in the proof of Proposition 5.1, and the Lean verification are available at `https://github.com/yspennstate/flow-parameter-diameter`.

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